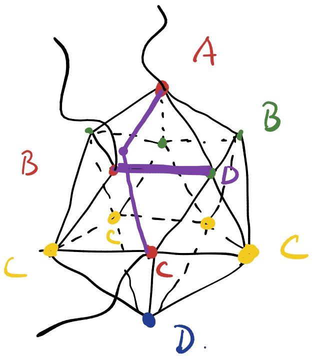
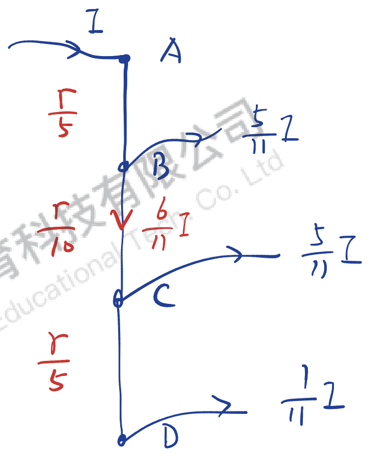
题7. (1) 伏安法. ①从A入I. 从余下每个出$\frac{I}{11}$.
②从C出I. 从余下每个入$\frac{I}{11}$.



⇒



$$\Rightarrow U_{AC1}=\frac{r}{5}I+\frac{r}{10}\cdot\frac{6}{11}I.$$

$$=\cdots$$

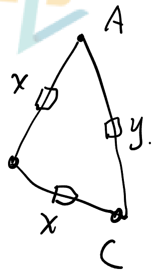
$$\Rightarrow R_{AC1}=\frac{2U_{AC1}}{\frac{12}{11}I}\Rightarrow\frac{7}{15}r.$$

(2) $R_{AC1}=R_{AC2}$ 并 $2r$.

$$\Rightarrow R_{AC2}=\frac{14}{23}r.$$

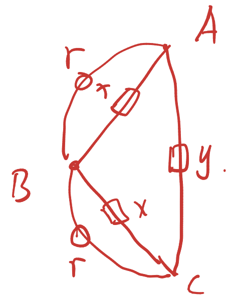
(3). 去掉AB、BC之后.

看. A、B、C之三端网络.



$\Rightarrow R_{AC2}=2x$ 并 y

没拆任何度时.

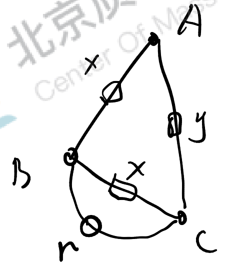


⇒

$R_{AB}=(r$并$x)$并$[(r$并$x)$串$y]$.

$$\begin{cases}x=\cdots\\ y=\cdots\end{cases}$$

(3)问:



$\Rightarrow R_{AC3}=\cdots$ ✓

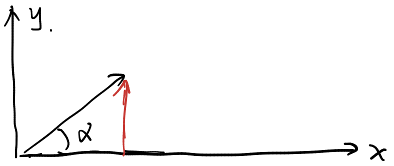
题6. 答案

$$(2)\quad x_0=\frac{U_0R_0}{\omega Bl\sqrt{(R_0-\omega^2L_0C_0R_0)^2+\omega^2L_0^2}}$$

题8. 答案: $\frac{\delta\nu}{\nu}=2.72\times10^{-4}$

34套.

题1.



$$e = \frac{|V_{y后}|}{|V_{y前}|}.$$

看. 碰n次之后. V_x 变为0.

若 碰后. $V_x > 0$. 则. 碰时. $\Rightarrow f = \mu N$.

$\Rightarrow$. ①碰后：

$$I_y = \int N \cdot dt = m \cdot V_0 \cdot \sin\alpha \cdot (1+e).$$

$\updownarrow$

$\uparrow e \cdot V_0 \cdot \sin\alpha$.

$V_0 \cdot \sin\alpha \downarrow$

$$\Rightarrow I_x = \int f \cdot dt = \mu \int N \cdot dt.$$

碰后： $V_{x1} = V_0 \cdot \cos\alpha - \mu \cdot V_0 \cdot \sin\alpha (1+e)$.

②碰后：. . . . $\Rightarrow$

$$V_{x2} = [V_0 \cdot \cos\alpha - \mu \cdot V_0 \cdot \sin\alpha (1+e)] - \mu \cdot V_0 \cdot \sin\alpha (1+e) \cdot e.$$

⋮

第i次碰后. (若 $V_{xi} > 0$).

$$V_{xi} = V_0 \cdot \cos\alpha - \sum_{j=1}^{i-1} \mu \cdot V_0 \cdot \sin\alpha (1+e) \cdot e^j$$

$$= V_0 \cdot \cos\alpha - \mu \cdot V_0 \cdot \sin\alpha \cdot (1+e) \cdot \frac{1-e^i}{1-e}$$

$\Rightarrow$ (一). 当 $i \to \infty$. V_{xi} 仍： $V_{xi} > 0$.

要求： $V_0 \cdot \cos\alpha - \mu \cdot V_0 \cdot \sin\alpha (1+e) \cdot \frac{1}{1-e} > 0$.

$$\cot\alpha > \frac{\mu(1+e)}{1-e}.$$

位移由2部分组成：

①. 上下弹时. $\Rightarrow$ $t_0 = \frac{2 \cdot V_0 \cdot \sin\alpha}{g}$.

$\Rightarrow$ 位移： $\Delta x_1 = V_{x0} \cdot \frac{2V_0 \cdot \sin\alpha}{g}$

$$\Rightarrow t_i = e^i \cdot \frac{2 \cdot V_0 \cdot \sin\alpha}{g}.$$

$$\Rightarrow \Delta x_i = V_{xi} \cdot e^i \cdot \frac{2V_0 \cdot \sin\alpha}{g}$$

$$= \left(V_0 \cdot \cos\alpha - \mu \cdot V_0 \cdot \sin\alpha \cdot \frac{1+e}{1-e} \cdot (1-e^i)\right) \cdot e^i \cdot \frac{2V_0 \cdot \sin\alpha}{g}$$

$$\Rightarrow L_1 = \sum_{i=0}^{\infty} \Delta x_i.$$ ← 等比求和.

L_2： $V_y \to 0$. $V_x > 0$ 时： 纯滑；

$$L_2 = \frac{V_{x\infty}^2}{2 \cdot \mu g}$$

$$= \frac{\left(V_0 \cdot \cos\alpha - \mu \cdot V_0 \cdot \sin\alpha \cdot \frac{1+e}{1-e}\right)^2}{2\mu g}.$$

(二). 当第n次碰后. 速度为0：

$$V_{x(n-1)} > 0 \geqslant V_{xn}.$$

$$V_0 \cdot \cos\alpha - \mu \cdot V_0 \cdot \sin\alpha \frac{1+e}{1-e}(1-e^{n-1}) > 0 \geqslant V_0 \cdot \cos\alpha - \mu \cdot V_0 \cdot \sin\alpha \cdot \frac{1+e}{1-e}(1-e^n).$$

$$\frac{1+e}{1-e}(1-e^{n-1}) < \frac{1}{\mu \cdot \tan\alpha} \leqslant \frac{1+e}{1-e}(1-e^n)$$

$$1-e^{n-1} < \frac{1-e}{(1+e) \cdot \mu(\tan\alpha)} \leqslant 1-e^n.$$

$$\Rightarrow e^{n-1} > 1 - \frac{1-e}{(1+e) \cdot \mu \tan\alpha} \geqslant e^n.$$

e: 恢复系数

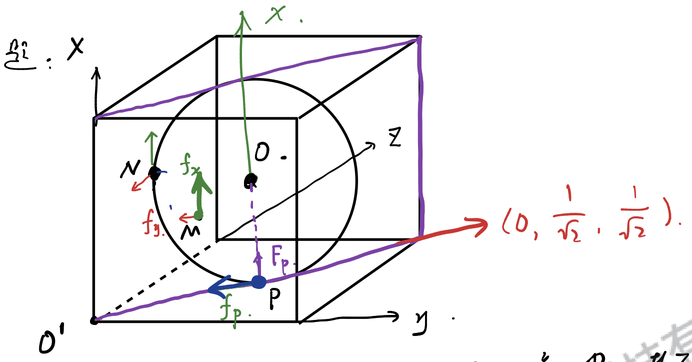
$$\Rightarrow n = \left\lceil \frac{\log\left(1 - \frac{1-e}{(1+e) \cdot \mu \tan\alpha}\right)}{\log e} \right\rceil$$ ← 向上取整

$\ln e$.

$\lceil 1.01 \rceil = 2$.

$$\Rightarrow L = \sum_{i=0}^{n-1} \Delta x_i.$$

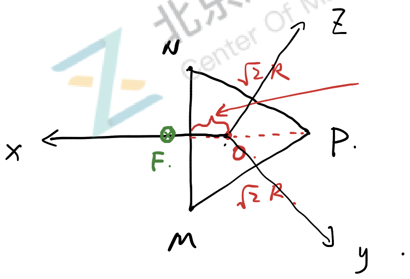
2题：



N: x-z 面. 切点. M: x-y 面.切点. P: yz面 切点.

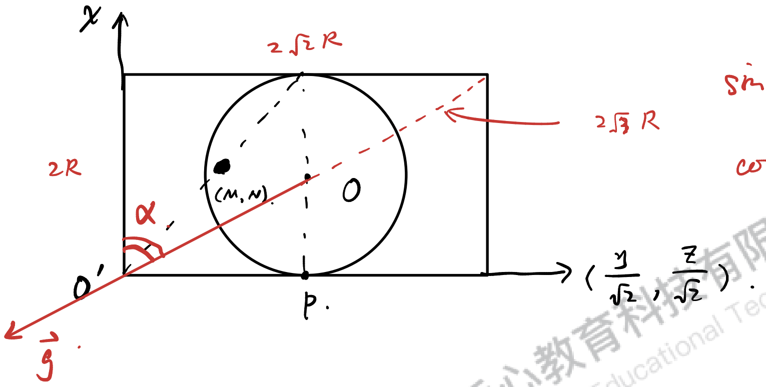
$\vec{g} \parallel OO'$ $\vec{g} = (-\frac{1}{\sqrt3}g, -\frac{1}{\sqrt3}g, -\frac{1}{\sqrt3}g)$.

俯视图：



$\frac{1}{3}\cdot\frac{\sqrt3}{2}\cdot\sqrt2 = \frac{1}{\sqrt6}\cdot R$.

取：x 与 $\frac{\hat{y}}{\sqrt2}+\frac{1}{\sqrt2}\hat{z}$ 相交的面：即 MN ⊥ 此面.



$\sin\alpha = \frac{\sqrt2}{\sqrt3}$.

$\cos\alpha = \frac{1}{\sqrt3}$.

情况一、滚：以 MN 为轴. 力矩：

$$F\cdot(\lambda-\frac{1}{\sqrt6})R = G\cdot\frac{1}{\sqrt6}R$$

$$\Rightarrow F = \frac{G}{\sqrt6\cdot\lambda-1}. \quad ①$$

$\frac{1}{\sqrt6} < \lambda < 1$. ← ⊛

滚动且是不打滑： $f \le \mu N$.

由对称. M: f_{Mx}, f_{My}. N: f_{Nx}, f_{Nz}. ⇒ 令 $\begin{cases} f_{Mx} = f_{Nx} = f_x. \\ f_{My} = f_{Nz} = f_y. \end{cases}$

以O为矩心： $\lambda R F = 2\cdot\frac{R}{\sqrt2}f_x$.

$$\Rightarrow f_x = \frac{\lambda}{\sqrt2}F. \quad ②$$

沿 $(0, \frac{1}{\sqrt2}, \frac{1}{\sqrt2})$ 方. 力平衡. 设 M. N. 压力: F_N

$$(G+F)\cdot\sin\alpha = F_N\cdot\frac{\sqrt2}{2}\times2 - f_y\cdot\frac{\sqrt2}{2}\times2. \quad ③$$

$$\Rightarrow N = f_y + \frac{1}{\sqrt3}(G+F).$$

$$= f_y + \frac{1}{\sqrt3}\cdot\frac{\sqrt6\lambda\cdot G}{\sqrt6\lambda-1} = f_y + \sqrt2\lambda\cdot F.$$

⇒ 不滑要求： $\mu \ge \frac{f}{N} = \frac{\sqrt{f_x^2+f_y^2}}{f_y+\sqrt2\lambda F} = \frac{\sqrt{(\frac{\lambda}{\sqrt2}F)^2+f_y^2}}{f_y+\sqrt2\lambda F}$

不妨设. $f_y = \xi\cdot\lambda F \Rightarrow \mu \ge \frac{\sqrt{\frac12+\xi^2}}{\xi+\sqrt2} \Rightarrow \cdots$

$\Rightarrow \mu \ge \frac{1}{\sqrt5}$. ← 与用摩擦角的结果一致

$\lambda > \frac{1}{\sqrt6}$. $\mu \ge \frac{1}{\sqrt5}$：以静摩擦为前提计算的.

若以动摩擦为前提. ⇒ M: $f_x \ne 0$. $f_y = 0$.
M: $f_x \ne 0$. $f_{Nz} = 0$.

⇒ 代与方程③：

$$(G+F)\cdot\sin\alpha = F_N\cdot\frac{\sqrt2}{2}\times2$$

$$\Rightarrow F_N = \frac{\sin\alpha}{\sqrt2}(G+F) = \sqrt2\lambda\cdot F$$

$$\Rightarrow \mu \ge \frac{f}{N} = \frac{\frac{1}{\sqrt2}\lambda F}{\sqrt2\lambda F} = \frac12.$$

$\lambda > \frac{1}{\sqrt6}$, $\mu \ge \frac12$：以动摩擦为前提.的结果.

②. 滑： 过O. 且 // MO 的轴. 力矩：

$$\lambda\cdot R\cdot F - 2\cdot f_x\frac{R}{\sqrt2} - f_p\cdot R = 0.$$

沿： $(0, \frac{1}{\sqrt2}, \frac{1}{\sqrt2})$ 方向. 力平衡：

$$(G+F)\cdot\sin\alpha - F_N\cdot\frac{\sqrt2}{2}\times2 - f_p = 0.$$

动摩擦： $f_x = \mu\cdot F_N$. $f_p = \mu\cdot F_p$.

$$\Rightarrow f_p = \frac{\mu\sqrt2(1-2\mu)}{\sqrt3(\sqrt2+2\mu^2)}(G+F).$$

要求 $f_p \geq 0$. ⇒ $\mu \leq \frac{1}{2}$ 才行.

再列一个方程:

$\vec{y}$方向平衡:

$$G+F = F_N\cdot\cos\alpha\times 2 + F_p\cdot\cos\alpha$$

$$\underline{f_x\cdot\cos\alpha\times 2}-f_p\cdot\sin\alpha$$

$$\Rightarrow \lambda F = \frac{\sqrt{2}\mu(\sqrt{2}+\mu)+\mu\cdot\sqrt{2}(1-2\mu)}{\sqrt{6}+2\sqrt{3}\cdot\mu^2}\cdot(G+F).$$

临界时: $F\to\infty$.

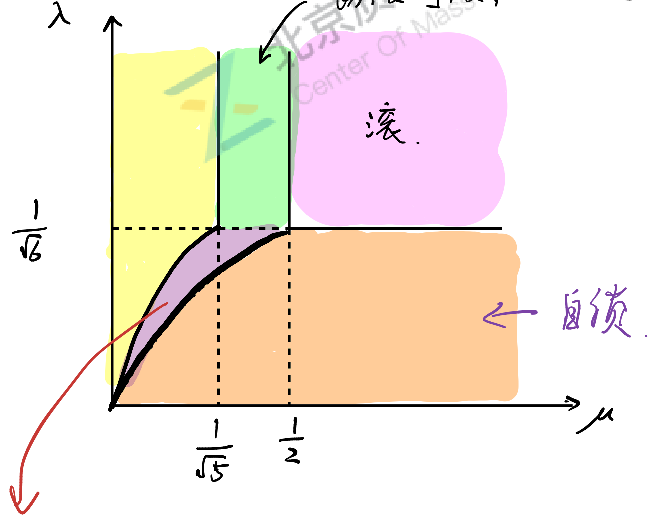
⇒ 临界时: $\lambda = \dfrac{\mu(1+\sqrt{2}-\mu)}{\sqrt{3}\cdot(1+\sqrt{2}\cdot\mu^2)}$

$\lambda > \dfrac{\mu(1+\sqrt{2}-\mu)}{\sqrt{3}(1+\sqrt{2}\cdot\mu^2)}$, $\mu\leq\frac{1}{2}$: 以动为前提.

∴ 滑动条件.

以静为前提. 算. 滑动条件.

动滚则滚, 动滑则滑.



动滑则滑. 动静则静.

题3: (1). $t_1=\frac{2l}{v}$. $t_2=\frac{3l}{v}$. $t_3=\frac{7l}{2v}$.

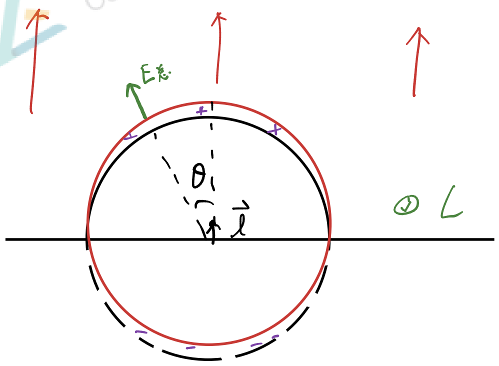
(2) ① 直达: $\frac{3h}{4v}$.

② : 先上岸. 再过去: $t_2=\frac{h}{2v}+\frac{2\sqrt{3}\cdot l}{3v}$

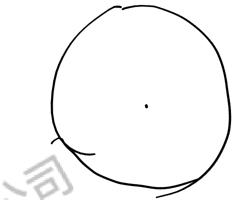
$t_1\leq t_2$ 即 $h<\frac{8\sqrt{3}}{3}l$ 时. ⇒ 直达

$t_1>t_2$. ⇒ 先上岸.

题4:



求 E_0=? 拉起来.



猜. 错位圆柱. ⇒ 验证.

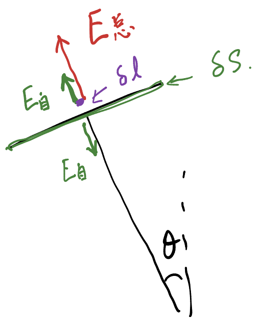
$\pm\rho_e$. 错开: l.

⇒. $\vec{E}_内 = -\frac{\rho_e}{2\varepsilon_0}\cdot\vec{l}$

导体电平衡 ⇒ $\vec{E}_内+\vec{E}_0=0$. ⇒ $\frac{\rho_e\cdot l}{2\varepsilon_0}=E_0$. ⊛.

受力: 只能看外场之电场 ⇒ 受力.

θ处 面电荷密度: $\sigma_{(\theta)}=\rho_e\cdot l\cdot\cos\theta=2\varepsilon_0\cdot E_0\cdot\cos\theta$. ②.



由高斯: $E_总\cdot ds=\frac{\sigma_{(\theta)}\cdot ds}{\varepsilon_0}$

⇒ $E_总=\frac{\sigma_{(\theta)}}{\varepsilon_0}$

$\boxed{取\ \delta l \ll \sqrt{\delta s} \ll R.}$

⇒ $E_自=\frac{\sigma_{(\theta)}}{2\varepsilon_0}$

⇒ $E_外=E_总-E_自=\frac{\sigma_{(\theta)}}{2\varepsilon_0}$

↕ 它也包括. 其余表面电荷

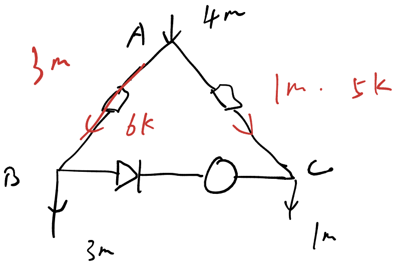
↑ 内力, 积分会抵消.

$E_0=\frac{1}{4}\sqrt{3\pi\rho g\cdot R/\varepsilon_0}$.

⇒ $dF=\sigma_{(\theta)}\cdot ds\cdot E_外\cdot\cos\theta$

⇒ $F=\int dF=\dots=\frac{8}{3}\varepsilon_0\cdot E_0^2\cdot L\cdot R.$ = 重力 ⇒

5. (1). 假设 BC有电流.



⋮

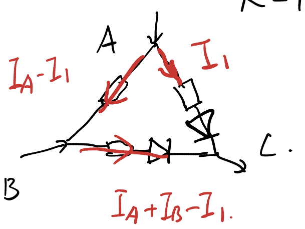
⇒ 矛盾.

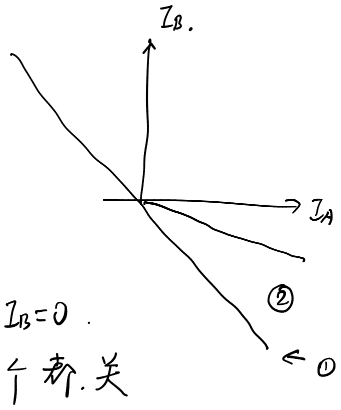
假设. BC无电流： ⇒ $\varphi_C > \varphi_B$ ⇒ 假设成立

(2) 假设二极管导通. ⇒ $I_1 = 2mA.\ I_2 = 2mA.\ I_3 = 1mA.$

⇒ $\Delta U = 2V.$

(3). $R = 1k\Omega$





⇒ $I_A + I_B = 0$.

① . 设 $I_1 = I_A + I_B - I_1 = 0$. 即 2个都关

⇒ $U_{AC} < 2V.$ $U_B < 2V.$

② . $I_1 = 0.$ $U_{AC} = I_A + (I_A + I_B) + 2 < 0$

⇒ $2I_A + I_B < 0$.

$U_{BC} = (I_A + I_B) + 2 > 2$.

⇒ $I_A + I_B > 0$.

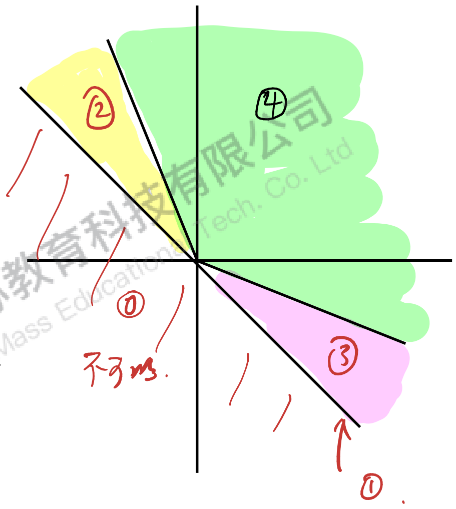
$$\Rightarrow \begin{cases} I_B > -I_A. \\ I_B < -2I_A. \end{cases}$$

③ $I_1 = I_A + I_B.$ ⇒ $I_A + I_B - I_1 = 0$ 即 BC不通

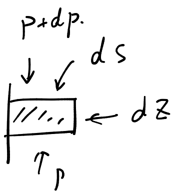
$$\Rightarrow \begin{cases} I_B > -I_A \\ I_B < -\frac{I_A}{2}. \end{cases}$$

④ AC、BC 都通：

$I_B > -\frac{1}{2} I_A$.



题6. 计算中 T、P 变化均为足小量：展开到一阶.



绝热大气：

$dp = -\rho g \cdot dZ.$

$\rho = \frac{\mu p.}{RT}$ μ：摩尔质量.

$p^{\gamma-1} \cdot T^{-1} = 常.$ $\gamma = \frac{7}{5}.$

⇒ $(\gamma-1) \cdot T dp + (-1) \cdot p \cdot dT = 0.$

⇒ $P_{(Z)} = P_0 \cdot \left(1 - \frac{2\mu g Z}{7RT_0}\right)^{\frac{7}{2}} \approx P_0 \cdot \left(1 - \frac{\mu g Z}{RT_0}\right).$

$T_{(Z)} \approx T_0 - \frac{2\mu g Z}{7R}.$

对气球：力平衡： $m_0 \cdot g + \rho_{内} \cdot g \cdot V = \rho_{外} \cdot g \cdot V.$

热平衡： $P_{热} = 4\pi r^2 \cdot K \cdot (T_{内} - T_{外}).$

$\nearrow$ 热功率.

内外压强一样：内部： $P_{外} \cdot V = \nu_{内气} \cdot R \cdot T_{内}.$ ↗ 状态

⇒ $P_{外} = \frac{\rho}{\mu} \cdot RT_{内}.$

⇒ …

$$h = \frac{\frac{1}{T_0} - \frac{1}{T_0 + \frac{P}{4\pi r^2 K}} - \frac{3m_0 \cdot R}{4\pi r^3 \mu \cdot P_0.}}{\frac{5\mu g}{7R} \cdot \left(\frac{1}{T_0^2} + \frac{1}{T_0\left(T_0 + \frac{P}{4\pi r^2 K}\right)} + \frac{2}{7\left(T_0 + \frac{P}{4\pi r^2 K}\right)^2}\right)}$$

题7. (1). $f_0 = \frac{n_2 \cdot R_1}{n_2 - n_1}$

(2) $R_2 = 80\ cm.$

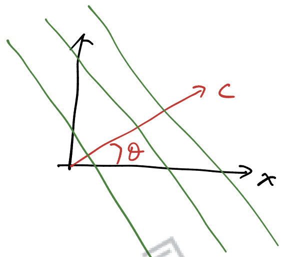
(3). $f = -\frac{R_3}{2}.$

(4). $u = \frac{320}{9}\ cm$ $v = -\frac{320}{3}\ cm$

(5). $u = \frac{320}{7}.$ $v = -320.$

横向：纵向放大率之比. 都是1.

题8. 不慌. 真空中.



$$E = E_0\cos(\omega t - k_x\cdot x - k_y\cdot y + \phi_0).$$

$$k_x = \frac{\omega}{c}\cdot\cos\theta. \quad k_y = \frac{\omega}{c}\sin\theta$$

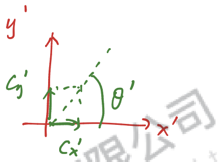
地系中: $\phi=\phi_0$: $\omega t - k_x\cdot x - k_y\cdot y + \phi_0 = \phi_0$

$$\Rightarrow y = -\frac{k_x}{k_y}\cdot x + \frac{\omega t}{k_y} = -\cot(\theta)\cdot x + \frac{c\cdot t}{\sin\theta} \quad ⊛$$

换到. u向右之系中.
$$\begin{cases} x = \gamma(x' + u\cdot t') \\ y = y' \\ t = \gamma(ct' + \frac{ux'}{c^2}). \end{cases} \quad \gamma = \frac{1}{\sqrt{1-u^2/c^2}}$$

代入⊛ ⇒ $y' = -\frac{\gamma(c\cdot\cos\theta - u)}{c\cdot\sin\theta}\cdot x' + \frac{\gamma\cdot(c - u\cdot\cos\theta)}{\sin\theta}\cdot t'.$

速度变换:
$$\begin{cases} c_x' = \dfrac{c\cdot\cos\theta - u}{1 - \frac{u\cdot\cos\theta}{c}} \\ c_y' = \dfrac{c\cdot\sin\theta}{\gamma(1 - \frac{u\cdot\cos\theta}{c})} \end{cases}$$

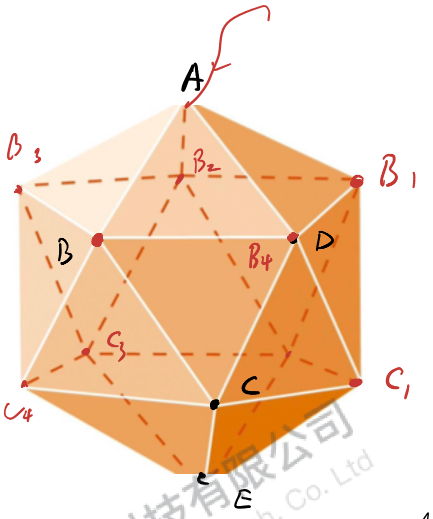


速度: $\tan\theta' = \frac{c_y'}{c_x'} = \frac{c\cdot\sin\theta.}{\gamma(c\cdot\cos\theta - u)}$

⇒ 斜率相乘: −1 ⇒ 相互垂直

33套. 7题.

(1). 完整之正二十面体.



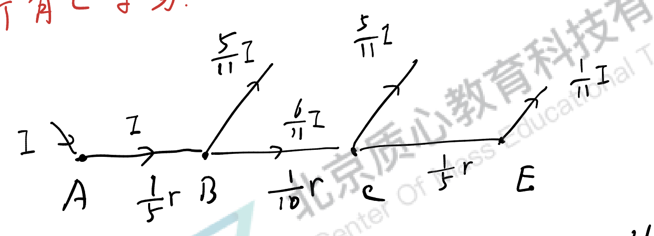
① 从A入. I. 余下出 $\frac{I}{11}$.

② 从C出I. 余下入. $\frac{I}{11}$.

所有B等势. ⇒连

所有C等势. ⇒连

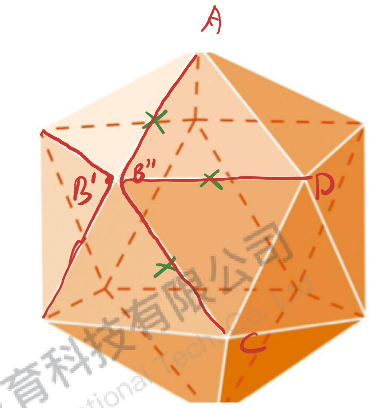
⇒



⇒ $U_{AC1} = \ldots$ ⇒ $R_{AC} = \frac{2\cdot U_{AC1}}{\frac{12}{11}I} = \frac{7}{15}r.$

同理. (2)中如是.

(2) 求 R_{AC2}



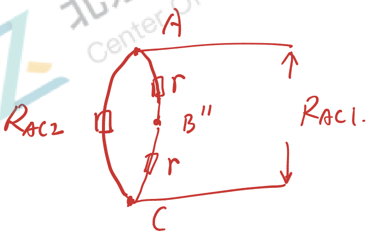
⇒只去掉AB、BC一致

⇒(1)对比.

B处可以. 切分.

切分之后 B', B'' 仍等势 ⇒ 可以切.

⇒



⇒ $R_{AC1} = R_{AC2}$ 并 $2r$ ⇒ $\frac{7}{15}r = \frac{R_{AC2}\times 2r}{R_{AC2}+2r}$

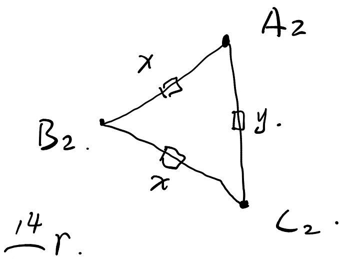
⇒ $R_{AC2} = \frac{14}{23}r.$

(3) 只去AB. 求 R_{AC3}.

设: 只去掉AB、BC之正二十面体.

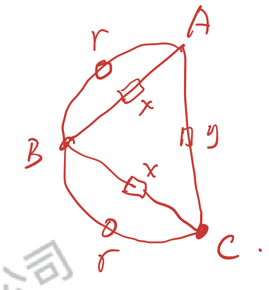
取. A、B、C三个点. 有接线柱

⇒三端网络.



R_{AC2} = $2x$ 并 y = $\frac{14}{23}r.$

补全成正二十面体. 完全相同.



R_{AB1} = (r并x) 并((r并x)串y).

第1问中: { A入. 全出. / B出. 全入

$x=r,\ y=\frac{7}{8}r.$

⇒ $R_{AC3} = \frac{21}{38}r.$

$$\Rightarrow R_{AB1} = \frac{\frac{1}{5}Ir\times 2}{\frac{12}{11}I} = \frac{2}{5}\times\frac{11}{12} = \frac{11}{30}\cdot r.$$

法二: 暴力展开 = 很打击.